

ActInf Livestream #020.1 ~ The Emperor's New Markov Blankets (full upload)

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hello and welcome to actin flab this is the active inference live stream number 20.1 on april 20th 2021 so a lot of repeating digits and i'm looking forward to this conversation with all of you so thanks for joining welcome to the active inference lab everyone we're a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links that are on this slide this is recorded in an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams like muting when we're not talking and raising our hands so we can hear from everyone on the stack hello greetings today in number 20.1 we're in the first of two discussions that we're going to be having as a group regarding this paper the emperor's new markov blankets and so the goal of these discussions is to learn discuss be curious and find out how we got here and where we're going and why we're doing that and we're just going to be having a nice conversation with the authors which are we're just really appreciative that they've joined so we're gonna just go to the introductions and warm up and then we will just be able to jump around and see where the conversation takes us so i'm daniel and i'm in california i will pass it to shannon hey i'm shannon i'm usually in california but i'm in south dakota since the pandemic are we going straight into warm-up questions or just names in around the room you can say hello or give a warm-up thought okay um yeah so i'm excited today i'm excited to explore whether or not we can use markov blankets to describe groups of people and social dynamics and i tried to do this in a project before and abandoned it pretty quickly and reading this paper i think if i had taken the pearl blanket formulation which i'm sure we'll get into talking about later it would have been a lot more helpful so i'm excited to discuss that today um i'll pass that to steven hello steven steven here i'm in toronto and um it's nice to be here and thanks a lot for this paper because i think it's going to be really interesting to have all these bits of the jigsaw put together and i'm actually quite interested in the paper because i'm trying to work out when to

use the concepts of like the markov blanket to help describe processes and when just to maybe talk about hidden states and not focus so much on the the blanket part so that's quite useful for me and i will pass it over to dave i'm dave i am in baguio city oh 120 miles north of manila it's the blazing summer time we're a mile high city so it might get as high as 80 degrees fahrenheit i'm retired from information technology and my background is in the cybernetics of learning and teaching great and we have two authors here so maybe first we'll go to chris and then yela and just feel free to introduce yourself the project the collaboration however you'd like thank you so thanks a lot for inviting us i'm christolega i'm currently a postdoc at the rural universitat in bochum germany i'm joining from berlin though and i'm currently working on a research grant on bayesian cognition and and the conspiracy theories i'm specifically interested in how proponents of conspiracy theories discount new evidence that contradicts their beliefs and this paper from my perspective grew out of a really long i guess friendship with yellow we can say that we've met a long time ago at a conference in the netherlands in 2014 i believe on bayesian cognition and we were both fascinated by this new uh theory framework of predictive processing and i remember that already there there was a seed of some kind of doubt in me or or kind of a worry planted because it was a it was a conference on probabilistic approaches to cognition everyone there was supposed to be a bayesian in one way or another and suddenly it turned out that basically no one agrees with each other and everyone uses the terms in a different way and i feel like this paper is in a way um contextualizing that experience for for us but yeah sorry i'll let yella speak now okay uh great samuel abraham um i work at macquarie university in sydney but then also due to pandemic reasons located in amsterdam uh the netherlands um and so my my new research project is on the philosophical foundations of the attention economy but my phd was really in in uh in on a free energy principle and an embodied cognition um and indeed this this paper grew out of a group of people who are or colleagues and friends and and and acquainted with each other's work all roughly on on predictive processing and and free energy principle and active inference but from slightly different uh uh directions uh and i think it was sparked by kind of combination of fmsa kind of curiosity and skepticism about some of the developments that that were that were happening and have been happening already since a long time since since i guess 2013 or 14 when some of these papers originally came out and kind of slowly got taken up in more and more heavy philosophical projects and i think the more and more we read those papers the more and more we grew interested and and skeptical at the same time and uh this was really a project to try to finally kind of uh uh nail down what what was bugging us and and how perhaps how it might

be salvaged or or or what to do um so i'm happy this is now pre-printed i'm very happy we get to discuss this i was this morning doing the dishes and on my headphones listening to your exposition of of my paper and that was just a very great and mind-blowing uh experience where somebody explicated those ideas and not being one of the artists i think that's that's on the other all of us that's that's just fantastic to to have happen i think actually there is a third author latent uh there um so i'll try to pass on the baton to manuel and see if he does anything or responds otherwise we discontinue well hello everyone uh i'm just here trying to look around for a little bit it's uh past 11 p.m in tokyo where i'm right now so yeah i'll try to you know stay um along for a bit but yeah clearly i'll leave the other two guys like who are much fresher than me right now to to lead the discussion um as for me i i have more of a background i guess in computer science some maths here and there now i'm doing some work on control theory and estimation theory applied to neuroscience a lot of methods uh probably right now and which is the main reason i guess i started working on active inference and related ideas and yeah as the other two guys have already um said you know this paper was kind of born out of you know around our own curiosity and you know questions regarding like what the framework actually does and what it can do and what it cannot do so you know then i'll leave yellow to you know leave the discussion from here on awesome thanks everyone for joining it's great to have just the authors and and kind of read between the citations and then actually enliven between the citations so that we can ask questions that were maybe alluded to and hear your fresh perspectives what kind of feedback you've gotten on the paper etc so yeah yeah where would you like to start otherwise there's just a bunch of places we can jump in great um i'll just briefly uh give some introductory uh uh comments uh uh i'll keep it brief because i think we're all roughly on the same page having read the paper um so as i was saying uh this this project got started out from a combination of curiosity and skepticism about the some new applications of active inference and free energy principle and so i think as i understood it before it's it was a kind of functional lens to try to understand cognitive systems and in that sense offering a kind of unifying perspective and also it's a different cognitive constructs in the cognitive sciences and i was already kind of very interested in that and what we've what what we've seen in the in the last few years i think are one applications or at least attempts to apply it to kind of the harder questions so the hard question of consciousness intentionality etc which is more than this kind of functional lens an application to systems that are not necessarily cognitive so living systems as a whole or even like a global global systems or social systems and then as well using this terminology not only to as a functional lens to to

understand systems that you already deem cognitive but also to use those tools themselves to do democratic work uh so to decide what is a living or another living system what is a cognitive or another cognitive system so to somehow employ those tools of active influence onto these kind of questions um and i think we were yeah we were all interested in in in a set of questions and and seeing how that would uh uh uh um how that could work and whether those arguments really uh really hold um and i think to do that we needed to have a combination of uh one conceptual arguments uh two formal arguments and three philosophical uh uh analysis and and kind of combine those in the same paper and i think that makes it also made it a very challenging paper to to to write to see how all these different parts of the paper fit together and and perhaps it's also made it a challenging paper to to read but we'll probably get to that in a moment um and there's i mean there's something here i think that that kind of blocks the whole uh uh field uh some of us were on just uh a few hours ago on social media and there there's a new paper uh now uh among others by decoding polo evan thompson randall beer and i think one of the more meta discussions was was like how much mathematics how much conceptual work do you need to actually contribute to the more conceptual uh questions you want to to ask and i think that this is something that's intrinsically intentioned here within the literature that you kind of need to have a very technical understanding to to have to watch this question in the first place but then if you do that perhaps you lose out on the more conceptual issues or you you lose out also part of your readership because it's it's it's only a few people that understand it so there's something exciting there about fep as a whole and i think we tried at least to be um as accessible as possible in in writing this paper while at the same time supporting our arguments with with with with the formalism that we thought were were necessary and that uh ultimately brought us to to writing this and the thing in in the moment we have a lot of pre-prints that came out in the last few months all kind of trying to target and trying to to to to to criticize these these developments in any principle so i think some of you some of them you've discussed here already so from mel anderson papers thomas as a good paper on this as well along with the thompson paper so i'm really curious to see how in the next few months half years say this this debate will uh will will will point out i think there's a number of really interesting kind of proposals now and uh yeah also um keep checking this uh this lab to to see how you treat those other papers i'm really looking forward to that cool um very interesting blue do you want to just say hi give any introductory thoughts if you'd like sure good morning i'm blue knight i'm an independent research consultant from new mexico and i really enjoyed this paper um i don't know if you got to see like our

contextualization video but it was really good cool so i'll be kind of taking some notes everybody can also jump into the document and that was a big question that you raised there yellow um how much mathematics is required to understand or work with free energy principle and active inference and then which are the sort of skeletal elements and the sinews so to speak of the math and then which parts might be secondary or not up for some reason there might be a few seconds of uh of uh just flipping but yes sorry just yeah what um elements of the math are required to understand and work with these topics and then of course the major question of your paper and some of these other recent papers we've discussed is whatever the mathematics are what kinds of philosophical or metaphysical claims can we reach validly or or invalidly um yeah any other big questions or thoughts people can just raise their hand with a little bottom left icon so yeah we'll do steven and then blue yeah just one thing building on the bit what daniel said there is active inference in biological systems could be present as a way for life to function and evolve but there's also possibilities that the active influence process could create structures or networks which then form other systems which may not have to be so based on ergodicity in the same way it might be like you have the skeletal system which gives you locomotion so i suppose in some ways i'd be interested in that way that you know we're not tied to everything being active inference in terms of how we can understand its applications even though maybe at the core it has to be there at some level cool thanks anyone can give a thought on that or blue go for it so um just to kind of yeah to i don't know so stephen what you just said like active inference is present in biological systems i think you know we need to be careful before we pick a side over instrumentalism versus realism like is active inference present in a living maybe not maybe it's just the way that we use to represent a living system or to model a living system or like our best guess so far right i mean i think that the way that the science evolves um you know in terms of how we think about biological systems processing information will you know i mean active inference is what we have now but it probably won't be what we have in 100 years if we survive 100 years but but my question was uh you know in reading the paper i really it was brought to my attention and and steven's question so perfectly illustrates this so so to there's a great degree of semantics that's involved um and our choice of language i think really shapes how we um perceive the intentions of others so it's just is that is that the caution or is that the tale or is that the cd underbelly of the paper or to what role do you really think that language a lot of these things could be fixed with language yes cool yellow you want to go for it and then shannon is your hand great so i'll um in the those are two two great questions um and just to to clarify the

the last one um when you say uh language do you mean the kind of concepts that we are uh using or language as a whole as a system so i i mean like how steven just said systems have active inference like it's just like systems have active inference like i have a heart and a kidney and a liver but i don't know that i necessarily have active inference right so it's just our choice of of language in in that so not language as a whole but but language like carefully selected word choice could maybe prevent a lot of the the conflict um yeah i i i think so and then i guess generally you you might think that lots of philosophy is actually a kind of a linguistic therapy right you're just trying to very be very concise and using the right kind of terminology so that you don't slip into confusions that that then will cloud your thinking also uh so i think there's definitely some of that on the on the background in the in in this paper um but it's it's not only a language right there's also the the the formal models uh and and how to interpret those those those formal models um and there in in the translation between the technical use uh um of of of this terminology and the kind of everyday intuitions it gives us i think there is in this room for a lot of mistakes and slippage i mean the notion of blanket is one right i mean i have a blanket on my bed and that has that has a spatial dimension and so and you can be either on top of it or below it etc and and so when you then use the notion of a markov blanket this is the first intuition that people will will have uh and it turns out that actually no these kind of things don't have such a kind of existence right they don't they actually they don't they're not really spatial things in the world they're they are just aspects of mathematical model and so indeed in in kind of there the language is one but also just a particular way in which you understand your concepts either as everyday things or as technical things i think that there's a lot of those slippages there and i guess the whole literature is full with notions of model of inference and and then it's very unclear whether something is a inference it's a technical notion or inference as a philosophy of mind notion or inference is something else um so yeah there there are a lot of those those those uh things in the background and i think that that also uh points to the question that stephen was asking uh in terms of like we can think about the structures for locomotion and so like how does this relate to to active um and so this is this is i guess this is i mean perhaps this is a topic that's not really there in in in in in this paper uh so much but there there's a huge literature in active influence on how to understand the notion of model and whether agents are models of the environment and one way to see that is is thinking like well there is an an and an echo niche around this and we have through selection that might not might not be uh understood for active inference have been equipped with a particular kind of neural system particular kind of body that allows us to

to engage with it with that environment and in that sense kind of things like the the the structures that facilitate locomotion could be just as part just just as well be part of that model but then model in a very different sense as it's typically used in cooking designs um yeah so so that that's also that that's very much how you understand model uh uh really um shapes how you will answer these kind of questions about how active inference and locomotion also relate to each other thanks for that and um it's almost like the word semantic ironically carries a double meaning because people will dismiss a discussion well that's a semantic discussion on one sense they mean you're talking about words words words like you're kind of just saying random stuff we're debating about a word we're talking about the same thing it's semantic it's only semantic but semantics are meaning and so i i think i've blipped out or maybe for some people for a second but it's just so funny how semantic refers to both the big sense of meaning and the small sense of the word and we're all reducing our uncertainty by actually going into the details what is the markov versus the pearl versus the fristen blanket and can we just have a common reference point so that when somebody says i'm using frist and blanket in the sense of this then it will preempt some of these broader valid or sort of spurious questions so cool shannon and then anyone else who raises their hand yeah so i'll try to make this comment without too much rambling um but we talked with with mel um andrews with an s um hippolyta and thomas venice and we've been discussing like on what level the system implements active inference or is um has markov blankets in the system versus when it's useful for us to describe it that way so that realism versus instrumental debate and something that was came up in the conversation with an s something that's easy for me is to claim that there is no markov blanket necessarily in a group of people that are interacting if we look at clusters of fans in a stadium i might as a researcher identify this cluster of fans because they're interacting in a certain way with each other that's different from this other cluster of fans and some game event happens and that ends up being some sort of coupling parameter and now we can describe this as one um unified system one unified cluster of fans of people um and i am pretty happy with saying that there is not a real boundary in the world between group a group b or that emerges over um the larger group c but i really want to say that that happens in the brain and i really want to say that our like the neurons in our brains are interpreting the world as if there are boundaries and that's forming boundaries on neural processes um like there's this specific people are talking about manifolds a lot lately and there's this similar um fuzziness and what it means to be a neural manifold but like there's a certain manifold that's associated with observing this feature of the world and i wonder if you have any intuitions

from this distinction can we be realist about markov blankets or the fringe principle or active inference as applied to neural states i mean that's where these theories started while being anti-realist about applying these two dynamics in the world like groups of people great question also um just a technical note like i'm not sure what's happening with the active inference youtube or jitsi or whatever but clearly my internet is working because we're having the video call but the stream glitched so we can just continue not on live stream and then i'll just re-upload the full quality and it won't have any glitches so i'll figure it out for next week but it's all good it won't be um fully live but we can just still think of it the same way which is the vast majority of people viewing it anyway so yeah sorry about that i'll figure it out for next time but um thanks shannon for the question about the groups so anyone want to speak to the group question and about whether groups can be properly said to be having or uh modeled by these blankets or where where is that sorry david you so i mean daniel you you glitched for a second and uh yeah in the meanwhile yell asked if i want to answer the group question and i think i would actually i have some thoughts about this so i think one important thing about the free energy framework that especially how it's been developed recently is that people often like people infristance closed circle like ines hippolyta for example has caused the paper recently with him about markov blankets in which they talk about viewing different systems as agents so so they talk about a neuron as an agent right they talk about the whole organism as an agent and and it kind of scaffolds across multiple time scales and and spatial scales and i think our paper leaves it open to some extent leaves the question open whether there is some kind of scale on which if we fix the parameters of the model the models can be validated and they can produce new predictions or new new empirical findings in such a way where we would be forced to accept that perhaps on that particular scale there really are blankets in the fristen sense or at least that these models are so good at predicting certain events or findings or phenomena on a given scale that we would be inclined to give them a realist interpretation so i think this is still on the table and it's very much open it's just that given the ambitions and the scope of the free energy framework uh and the fact that they start with this assumption of viewing different systems on different scales as agents right now it's just not really clear um which level so to speak the this this way of framing phenomena would be validated in a realist context great um really what warrants or what license is what evidence would we need for realist interpretations and it's not even just a yes no because there's kind of a few kinds of realism as you pointed out like there's different notions of boundary and so a group might be very well said to have an informational boundary like this jitsi people who are not in the jitsi

are not part of this informational boundary but then in some other sense the group may not fall so neatly under the definition of a realist interpretation of a blanket um manuel yeah so uh i was a bit slow i guess after stephen and blue actually asked um a couple of questions earlier so i'm briefly gonna go back to that and then try to reconnect to what shannon uh was actually talking about so um as for i guess the the loose sense in which we might want to interpret active inference i think there is i don't know relatively low risk in just i don't know waving our hands and saying the systems do some sort of active inference as long as we know we keep it loose losing such a way that in my opinion you might just lose entirely like any meaning we're just saying well you know there's some active inference happening there in the same sense that there is some active inference happening in i don't know planets orbiting around stars or galaxies uh having some particular physical behavior and so we might end up just not having anything to say it's safe but it's also not very useful in a sense so in that direction i think that our work is trying to you know uh go in the direction where we want to say what active inference is not in a way it is important to say well you know this is what it is right now and this is what it cannot be and if you want it to be something else well you you have to tell me exactly what else you need for that thing to even work to begin with and then related to that there's also um so i think i might have heard this from alex shuntz before and i don't know who else before him might have mentioned um the idea of say an active inference coming from the i don't know highway and an active inference coming from a more normal if we can call it like that path the highway is what fristen has been trying to do more recently you know starting from first principles as he claims like from the physics of you know langevin equations and you know stochastic systems and then trying to say derive notions that are already established in non-equilibrium physics and from there get all the way to say a definition of what agents might be there is also say the the i don't know the normal way which i think is what uh shannon alluded to um you know at some point because you know this active inference theory was originally brought out in neuroscience with no ambition to be a first principle theory starting from you know uh launch of integrations that might describe any physical system in this universe it was just supposed to be some sort of helpful framework that in some way was going to generalize reinforcement learning which is an absolutely fair i guess uh goal and ambition but has also nothing to do with all of the physics that is more recently being plugged into active inference and one could safely go say the the second route and just forget about all the physics it might leave it i don't know to a different um i don't know to a different status but it would still be potentially useful and it would still be something that we might look forward

to and that i guess lead me to um leads me to um something that shannon was mentioning about manifolds and in some sense um you know the the question about you know are markov blankets real or not is similar to yes are manifolds real or not um and here you know i don't want to say um i'm not really sure you know how to to go about this in in the same way that i don't know physicists generally don't think too much about well you know is this energy landscape real or not is my you know is my ball actually trying to fall to the ground or is it just well you know it is a good metaphor it works really well we tend to use some you know a genitive language in there but i guess that if you hard press a physicist you will probably hardly get an answer about you know oh is this energy escape like real is there any an energy scape inside the ball that the ball is using to actually roll down the hill well probably no it is the kind of question that you know well i will still be using lagrange mechanics i will still be using emiletonia mechanics but i will not be you know opening up a ball looking for the energy landscape there in the same way i will still be using manifolds well personally because you know population codes are revealing like to to be pretty useful in neuroscience but i will not say like start looking inside the brain like well where's the manifold there if that makes sense yes totally makes sense and real and useful when they're aligned people celebrate and when they're not aligned there's often another story why something is still really useful even though it's just a useful fiction or something like that and i think that um vector alignment is really what we want to nuance we want to have like an hour hand and a second hand in a minute hand we want to be clear where are we just specifying a partially observable markov decision process and fitting data and then where are we in the interpretive frame and where are we in a metaphysical frame even beyond the um interpretive steven and then anyone else raising their hand i think as well as where we say if it's real or not there's the sense of at what scales and i like the fact that you talked about scales is what scales um and at those scales how much plausibility is there for certain types of inference processes to be possible i suppose that's where you know if there's no ergodicity present or seems to be very little at certain scales then maybe it's less likely that that mechanism is an active inference mechanism although there may be some elements of it so i think that that also changes the way that the realist aspect could be brought in it's like how much could it be a blanket aspect or maybe like you say that other scales unlike say a single cell which is diffusing chemicals over a membrane and there'd be some weak mixing process going on there's some um yeah there's some ways that variation or expected free energy are not the only mechanisms even suggesting that it's a mechanism is something that's mechanical it's these are what we want to understand

and these are the discussions and then are they equivalent and do mechanisms have to be real things or can a mechanism be something that doesn't exist in the world or a conceptual mechanism or could it be just a pattern that doesn't exist in one measurable thing we can put on a scale like a machine um so cool questions dave and then anyone else raising their hands okay uh sometimes the math is trusted too far and it's allowed to do a whole lot of work but sometimes the math gets crippled now i'm a student of gordon pasque and cyclicity and um self-stabilizing feedback is really critical to everything he did in learning theory and biology system science and what's the very first thing you do when you mention markov processes you stomp out the cycles you rip out those cycles you're just not allowed to have loops in a markov network well then they come back in because that's what people are looking for they're looking for self-stabilization and using the environment and the environment using you to enrich itself and then as you point out at the end of your paper well then dr fristin finds there's cycles in there and then he goes and rips them out all over again what has anybody ever just left the cycles in come up with a formalism that's richer than a markovian tree and then see if maybe things get easier and simpler great question dave and i think the big question is what are the inputs what do we want to get out of this math are we looking to make analytical claims or are we looking to have uh very general frameworks for control theory um or for estimation theory even like manuel mentioned so anyone can raise their hand but um chris uh thanks so i just wanted to follow up on the last two questions but i see that there's already a theme emerging in a way so one thing is about the maths and you already the question that was posed at the beginning how much the understanding of maths informs research that people can do into the predictive processing framework and i think and now we return to this question in a way with this question of of cycles and the fact that we actually analyze you know we leave the cycles out and i think it's in some ways useful to think about our paper in many ways and and what's happening with the free energy framework as a kind of a pyramid where you know formalisms mathematical formalisms are stacked one on another and finally the free energy principle really lies at the top and what we've been trying to do in some ways is to lead our readers you know through the steps going up to the free energy principle and like how it's supposed to work but the thing is that of course there are many different ways to get there so you know someone might have like you you can think about the cycles like the the the topological method that we took with directed uh graphs and the the the way of thinking about dynamical coupled systems and loops and stuff like that you can think about this as different slopes of the same pyramid right in the end the point is to that you want to get to the free

energy principle exactly you want to get to the free energy principle and the free energy principle lies on top of a lot of work done in many different formal fields and i think that i think that the the the last question asked by dave pointing to the possibility that perhaps if we leave the question of directed the cyclical graphs aside and just focus on cyclical systems perhaps we could arrive at a new con formal construct that could later turn out to be somewhat equivalent to to a mark of blankets but wouldn't be so easily conflated with mark of blankets i think that's a very interesting possibility uh that probably should be explored in the literature um and the other thing i wanted to just address very briefly is the fact that the other theme that seems to be emerging here in the discussion is this theme of patterns and and what's real and what's not real and i think here it's useful to bring back uh or or kind of points that dan dennett has already done some important work in the past on this with his work on real patterns and and what does it mean for a pattern to be real and what's the difference between a pattern being real or and not being real um one interesting thing if we think about active inference for example is that if we think about the social dynamics uh we can think for example about the dynamics in the army right a platoon clearly has some kind of a blanket like the platoon is trying to maintain its homeostasis they they are all trying to stay alive especially in a hostile environment far away from the equilibrium right the interesting thing about the example of a platoon if i and i hope i'm not getting that wrong now but i'm sure manuel would correct me if i do is that we don't actually assume that the the that the platoon would be statistically ergodic because it definitely shouldn't uh shouldn't visit some of its possible state spaces portions of the of its state space like the whole point of putting people into a platoon is so that they won't visit certain regions of the state space um so those are interesting questions here and of course the question about like are platoons uh you know are they uh it's an interesting it's an interesting question to ponder dan dennett previously has asked the question about money right so is the value of money real obviously it has direct impact on our lives we be it has a direct impact on how we behave uh in virtue of the kind of social construct and the social scaffolding uh on which it rests but we all kind of agree that you know the the existence of value of money is just a it's just some kind of a fiction that we all buy into especially now that the gold parity is no longer a thing nice cool points and and it's really remarkable how this what is real i mean in some sense it makes sense it's the classic philosophy question plato's cave and what's real and it's hard to differentiate it sometimes because it's used in very many senses so sometimes like the pyramid it's almost like we're surfacing it like an iceberg to mix metaphors and then people can visibly attack a formalism well the directed acyclic graph

doesn't represent reality because reality has cyclic elements and then we have to sort of change the formalism a little bit like could we run it in a sequential way could a message passing algorithm emulate a cyclic system even though locally computationally it's linearized but you lose some other attribute is that real then would the model be real so fun questions um yella and then stephen great um so i think a question kind of related slash orthogonal to uh to to to the real question is a question about uh what's in the paper called reification right so it's it's a fact that if you make a mathematical formalization of something that mathematical formalization might is quite often idealization of the target phenomenon that you're that you're investigating and then as long as you treat that mathematical structure as an idealization of the target phenomenon and don't automatically transfer the properties of the idealization to the to the target phenomenon then that's completely fine and then the uh where it gets upside down is if you suddenly take that mathematical formulization as somehow being at the root of the appearance of the target phenomenon that you're investigating so we make this case for for the kind of graphical model uh example where there is a complicated system you make a graphical model of it which is you know by all means a kind of simplification of that very system and then if you start to say look markov blankets were actually in that system you're also implying that no actually that mathematical idealization uh of of of of the system was actually already in the target system that you were that you're investigating and that's that's the thing else will move that you make the mathematics itself more real than you're than you're supposed to make it um and of course i think we said this towards the end of the paper you you can buy the bullet right and say like look no matt reality at this very core is mathematical so so so we're fine but that that is something that doesn't itself come from the formalism but requires actually a pretty strong metaphysical assumptions and so the question about in that sense the question about the reality of a mathematical structure is also intertwined with this question about reification about how do you see the relationship between the target phenomenon you're investigating and the mathematical formalization of the target phenomenon i think these two questions kind of go go go ahead and then thank you steven than anyone else yeah i think this this mathematical piece is important as well because i think there's often the assumption that we're always inferring this model of the world and how to adapt and interact with the world but one of the key things particularly higher order animals and humans like the platoon is at times they want to be disordered or this coherent self but ultimately when they're in the field yes they want to get into the most optimal ways to shoot and do things but they need to gain the system they need to not do what the other people think

they're going to and they have to guess what someone else might do which is non-ergodic something which is you're basically gaming each other's environment so that you're basically like prey and predator you're you're also then taking another stance so you step outside of what would be that normal dynamical free energy minimization process and that is sort of sitting on top of the other process so i think that's a kind of an interesting confusion cool dave than anyone else the question of what is or wasn't what isn't ergodic seems to interact a lot with scale and this is something that can be manipulated in an organization uh sometimes to what seem like contradictory or paradoxical effects for a long time the us military officer corps has had an up or out policy if you didn't get promoted within a certain amount of time you were invited to leave the military and go do something else with your life which sometimes didn't work well you started running into um the um uh oh you know the the the british sociologist from the 50s with the famous organ the most famous organizational law such and such as law yeah people are you know they end up in their their at their level of incompetence well much more recently just over 20 years ago the u.s department of justice started adopting an up or out policy which quite likely is a lot more toxic than having that in the military because the military has usually a fairly narrow kind of purpose if it's being used at all in the department of justice the fbi especially if you're a local investigator you're expected over a period of years not just to know your colleagues and the cops and the prosecutors and the reporters and the informants and the way the underworld works whether you're stomping them or not you're expected to develop an enormous amount of highly differentiated knowledge of south cincinnati or the east bronx well then in the late 90s somebody decided oh no wait a minute this has this leaves a possibility of local corruption we're allowing nests of incompetence and nests of bribery so we're going to require people to leave no matter how much you love working in the south bronx you either get promoted to washington or you get out so some the intention presumably was to keep a nice smooth pipeline a clear flowing stream but you end up requiring incompetence instead is there a larger scope at which you become scale independent and that sort of thing and oh yeah it looks muddy at a low level but it's actually ends up being clearer or is it just screwing up question there's a lot there one piece i kind of heard is like there's systems that we can actually control like a software program we can write a program with a strictly defined topology of variables and so it's easier to take a realist interpretation of a program where we can define all the pieces and we can know everything that's happening but once we're dealing with systems that we didn't directly control we know that we're dealing with a map territory distinction which i guess is something i'll flip to

because this was just a great in a very accessible point which is wherever we go down the rabbit hole with the math we have to keep that map territory embodied metaphor in mind when whether it's a phone you're looking at and it's deep learning and it's maps and recommendations or whether it's a hand-drawn map that your friend made or whether it's a historical painting it's a map and we can all see how that differs than the actual quote realism of the territory which isn't even like what we see it's actually something a little bit else so going uh and holding to this map territory distinction is kind of a useful thing and so it's great that you brought it up also on the sort of symbolic and evocative wavelength um what about the title or what inspired the title how did you see that story or that narrative play out because you say a lot with a few words i think we stumbled on upon the title and just thought it was too too good not to use but in terms of the uh what you have there oh goodness um i i i heard you uh uh uh uh explained the story during the the pre-recording which is very uh nice uh but i think there was one element there that you that you that you didn't mention and that's there that there are somehow these these these swindlers in this story fell to the rest of the the lackeys and so in the palace like these are a set of clothes that you can only that you can't that you can only see if you're very clever if you're if you if you're too dumb you won't be able to see these clothes so there is and all the lackeys and everybody involved in the story there there is a kind of doubt like is this me am i too stupid or is something fishy going on here and i think that kind of state of suspense of not really knowing i think that's perhaps the the key to the title uh we don't we well knowing to suggest that there were swindlers in this story or so but that kind of idea like hey am i am i too stupid to get this or is something weird going on and i think uh uh everybody on everybody working on free energy principle has or at least should have had that feeling to to to some extent uh so i think that's the key to the the title without giving away uh too much about the rest of the story yes shannon i somehow missed that nuance from from um but it seems really question for the new paper that just came out too and the twitter post response on it too is maybe certain technical aspects of the free energy principle weren't described with the specific mathematics and formalism that you need to properly criticize the free energy principle um which maybe maybe you do maybe you need to have that much detail to properly criticize it but if it's a principle that we want to use in our empirical science and our understanding of whatever subject matter we're looking at and also something that we want to use i i assume that in science we want to use this to also inform the world about something that we know about the system that we're studying um whether it's something like this is a true real fact of the system or this is the most useful way that we can understand the

system so that we can apply treatments you know maybe translate to applied clinical practice or something like that then at some point it needs to be possible to abstract away from the formalism without losing the detail or without losing the assumed detail that the formalism brings in and maybe it's just maybe like we have to work through this nitty gritty stuff and we have to fight over very small mathematical details and people that are better at math than i have to make that fight before like we can bring it in and people in applied practices and maybe even the greater public can start exactly what this means but i don't know when it when is the point that we can get there what is the point that we can say okay we're done arguing about the formalism we know there's subtle differences but i can still have a conversation with my friend who's just interested in my phd work but hasn't done a phd themselves or something like that i'm going to stop talking as i start trailing off with less points here but just throwing that out there thanks shannon cool points manuel so i hope this doesn't come along as a say negative um uh perspective in a way but i guess that beyond uh what shannon mentioned already as being part of the problems which are which are absolutely real like when do when can we stop like getting into the nitty-gritty stuff so that we can actually do some work on top of that and here i'm gonna break allowance like in um like for freezing like for once where i say that you know part of what makes this job difficult is also like the um the idea of dealing with definitions use the same words across different fields so we have an example here for uh with ergodicity so the ergodicity definition the physicist study is um yeah this idea of having a system that visits all of the spa all of the state in um uh in a state space such that you can replace time average with ensembl averages if you look at it from the perspective of markov processes which i think they've actually brought out at some point a periodicity actually becomes a part of the definition if you look at it from the perspective of ergotic theory which is this field born like 70 80 years ago and that actually is supposed to just be the uh to just deal with their girdicity then what they've actually suggested in their field is weak mixing so you can have systems that are not ergotic sorry that other gothic but not weak mixing um so all in all without going like too much into details i'm just trying to suggest that there is also a level of misunderstanding that comes from people having different backgrounds so on top of just say understanding what is going on we also need to have i don't know a paper or two that just list definitions as you know i'm using this definition from this specific field you can find it in this book [Music] different people might want to say well this is not really how i was taught this but this specific concept but at least it is something that i can engage with so a lot of the confusion especially on ergodicity actually comes out just of you know people having

different backgrounds none of them are wrong because all of them are just using different definitions unfortunately totally agreed with that it's almost like a scientific project what we're seeing is that there's an enormous desire to contribute to a shared scientific project and to make these ideas not just something that rests in the head of an expert but something in our exocortex in our informational niche something where we can all just see the affordances in a bigger project like wikipedia and improve it and then in service of we need um alignment on really the tools and the people and the ideas and the ideas are more on the philosophical side and the tools as we're seeing are more on the technical side and something that's right at the intersection of tools people and ideas are really the terms and their definitions and that pursuit of clarity as well as improved ability to translate and generate introductory material is really key and that active inference ontology type work is one of the things we're working on in actin flab because we totally agree when there are multiple definitions then it's either a don't ask don't tell policy on definitions or people do go into the details and then the paper is criticized for providing a specific detailed take or they make nine out of ten points but then it leaves um a shadow of doubt about the tenth of tenth point because it seems like it's going too technical and when people do a linear regression or a time series model even when lives are on the line no one says wait a minute i need the least squares error to be real but we're seeing criticisms about what's real here and so that's such a fascinating pattern that really will be explored i think in the coming months and years this line of research has brought us to some fundamentals and we are bringing up some new questions and connecting ideas and terms in new ways so it's fun to be a part of and um i hope that those who are listening can see that they're a part of this active conversation and it's not like they um the authors released this paper and now the conversation is over this is sort of their opening call and i hope that we can hear a few more pieces about what does that call entail for those who take it up um yellow yeah great um i just want to briefly follow up on on on shannon's point and uh saying that at least i think our aim is also to to uh um to kind of tease apart a little bit the kind of empirical research program of of active inference dcm and so and the more kind of what we in the paper in the end called the kind of most ambitious version of blanket oriented ontology and and show that uh you don't need to buy into that heavy duty metaphysical framework in order to do empirical interesting empirical work with active inference or with with dcm or so so that kind of association back and forth uh there might be a good fit uh but there there's i think there's very good reasons to just say or marker blankets stay in my cool toolkit as a scientist and i can use them and employ them every once in a while but uh i'm not gonna speculate

about how how that that my toolkit fits onto reality etc right that's that's a different ballpark but as having it in your toolkit you can still do uh basically all the signs that that that's already so i just want to be clear that at no point do we want to say like oh people should do empirical work differently or so but it's really like if you want to use this toolkit to dive into a different set of questions rather than the traditional empirical ones you can ask with questions about the democratic agents or or so that that might look a lot like the empirical one but actually are not really uh then you are confronted with a different set of questions namely about how does mathematics relate to reality and so but by no means do we want to kind of uh block that and i think i mean a long time ago i started physics right and then if you if you over bs you get discussions about are atoms real and so and these are kind of right these are the kind of questions that are just that are just funny to raise and some people have more or less patience with them uh but ben by no means do any of those questions block uh the actual research in in physics they're they're they're orthogonal to each other and i think the same holds for for for neuroscience at least to some extent if you if you want to interpret data or so you know use use a mathematical toolbox for that then that's completely metaphysically innocent in that so to say it's just this this transition that has happened that we that we have doubts about to take issue with cool thanks and anyone can raise their hand one thought on that is that the structure of the implementation of active models with a partially observable markov decision processes helps us separate out for example our priors about different variables our generative model our affordances and our preferences so using that model helps us distinguish for example our aspirational beliefs about how the world should be or what we'd prefer from the sort of reality on the ground with what our affordances are and almost at a higher level this is where this discussion is coming can we separate out our technological affordances from our preferences for example that an integrated theory of everything would be preferable that just makes a theory better for it to be more explanatory more predictive more universal more transferable across different systems those are preferences we might have over distributions of scientific theories but that's not quite the same thing as what are our affordances and we don't know all the affordances yet there's theories that haven't been invented there's frameworks that don't exist yet so how how do we make sense at the lower level and then also at sort of a higher meta scientific level have a discussion across different frameworks without trapezing into essentially unknowable metaphysical questions stephen and then anyone else yeah i'd be interested one of the it sort of relates a bit to what dave was saying there about feedback loops and we had in the last session we had casper hesp talking about some of the modeling and

there you have these kind of cyclical um iterative dynamics at different scales of how you know how fast or slow the kind of the churn of um observations and sensory states and maybe states of belief and such things so there's kind of a cyclical patterns sort of in there so just be curious how that cyclical pattern relates also to other types of feedback which is maybe a bit more dynamic um or just other types of feedback loops that you might think might need to be thought yeah um i see chris and manuel both with henries yeah am i on youtube go ahead uh manuel if you have if you want to say something directly to the last question then go ahead right so um my bad for actually not covering that before but the idea i guess that um dave previously brought out and now stephen um kindly uh reminded us about like the this idea of feedback loops uh there is so let's say uh um i would say that my background might be more from control theory in some sense so i do see the need of uh considering feedback loops um because you know there's the sort of like bread and butter in control theory there is also a sense however in which i guess usually in control theory we draw diagrams but they are kind of you know representations that will leave there as in like well this is the type of system i'm dealing with it is some sort of like picturesque uh say a view of what is actually happening out there we are not doing maths on the say like the the diagram that we are drawing in i guess what is happening with bayesian networks and why people are so much say inclined to use them uh is the is the fact that i guess we can actually use bezel network to do calculations so the the work by perl which we briefly mentioned in our paper clearly not giving enough justice to the entire say framework you know is something that can be used on these models themselves so in a way the the directionality there the unrolling of time the idea of an arrow of time there makes all of these calculations possible it is by no means uh a way to say that there will be um um say system that there will be different systems necessarily different systems at different times we may still choose to group certain variables and say well this is just how i don't know the environment is evolving over time or how uh an agent is evolving over time and we might just call that environment we might just call that agent that's fine i guess but when it comes to say do calculations with that graph if we just start drawing loops then time disappears and time is what makes a lot of these calculations actually work uh at least say what the algorithms that um you know dania also mentioned uh with message passing on bazel networks uh that is what makes them work like the fact that it is clearly there are also algorithms for cyclic uh networks but they are more limited they have some say uh they have some stronger assumptions so i'm rolling time i i think is not an incredibly crazy assumption or an incredibly crazy same method to use there it is probably yeah something quite

reasonable we are still saying there's a there's a there's an agent that talks to an environment and an environment that talks to an agent we're just trying to make that say interaction explicit as a process that happens over time yeah thanks for that that that feedback and i suppose one thing that the question that can come up is um and i'm not sure if the pearl blankets could do this or something that the frist and blankets gives you the possibilities if you're dealing with entropic information and noisy kind of data that's coming in an entropy can be like zero dimensional it doesn't actually need to have a dimension and you can extract from that whereas in other cases you you're always working with something which is like a channel coming in with like a dimensional data source which is somehow like saying something about the thing out there as opposed to something which can be completely untethered from the environment so i wonder you know how that that idea of being able to extract shannon entropy information might be a unique property the fristen blanket or whether that could be done in other ways um so uh i guess there are a couple of considerations there like exactly what would be extracting information from what uh because i mean this sounds a lot like the sort of like maxwell's demon type of scenario which um you know it's a well-studied field but it's also say possibly controversial in some sense because well in physics you don't need to ask certain questions you just well there's a demon and i don't and i'm not gonna tell you what this demon is i'm not gonna tell you how he does his thing it just does it and this is how you know things will happen um the the thing about you know extracting information in the blanket sense well here i want to walk say um a very fine line um by saying that the there are possibly issues with this definition of the friston blanket and they have been raised before and i think martin beale has pointed them out in a recent paper and fernando rosas has also mentioned them in his uh work on causa blankets as a way to fix i would say fristen blankets in the sense that the information that can be extracted in those blankets is not past information is just information at one specific time step because of this unrolling it makes calculations easier but the sort of conditional independence is uh in within a time step is also across time steps because of the markovian um property of uh of this processes and i want to make sure this is not something that i first raised this came from people like martin beale fernando rosas and probably in first place nathaniel virgo um so uh and again all the credit goes to them so this idea of extracting information is really feeble for now it is not uh it is not something that i would actually you know bank on in any way cool so it's almost like it's said to be a strength of the markov blanket or any kind of blanket concept that it's just one time slice at a time but you don't get to have the blanket and eat it too because you can't have the through time benefits and then say

that you're simplifying with something that's only reading one value at a time though there's also probably a lot of technical details too so chris and then anyone else raising their hand onion uh here we go sorry so that was a really nice exchange and i think it also ties things back to to your point about the map and the territory distinction because uh one thing that we have like i think one of the intentions we had uh at least definitely me and joe who unfortunately isn't here but is very interested in these topics is then is the philosophy of science behind the notion of a scientific model and one thing that we have to like in if we frame things in this way it becomes very interesting how we understand persistence formalism such as mark of blankets for example because uh in philosophy there's often this distinction made between the model the construal and the target system of the model right so the target system of the model is a phenomenon we're trying to investigate in the world the model itself is some kind of a formal construction usually usually described in some formal language and then the construal is kind of the interpretive link that connects the formal structure to the concrete real world phenomenon and one interesting thing about this is that the same formal structure the same model can be applied in different by different scientists successfully using different construals right and this also goes back to something that manuel mentioned before about the multiplicity of backgrounds from which people come to the free energy framework active influence framework and it's not unproductive i mean we have good examples in the history of science in the history of computer science specifically of people parting certain notions from physics specifically into computer science and making breakthroughs so one very nice example of this is the so-called icing model of ferromagnetism which has been ported by william little and john hopfield into the neural network domain where they basically used the same formal structure to and gave it a different construal right to model a different a different worldly phenomenon so hobfield networks and later boltzmann networks have been very successful for example in modeling memory at least back when they were introduced now we have much more sophisticated models of course but it's a nice example of this kind of cross-insemination between fields where people can actually take a formal structure or notion from one field and successfully make progress in a different domain but one thing that's important to keep in mind is that you know little or hopfield never never never never published their neural network model and said like look it's a thermodynamical model of memory or something like that because it's not i mean in a sense of course it is because it's the the structure that has been mapped onto the target phenomenon is the same as in the icing model or the spin glass model the the thing is though that the construals are very different and

i think it's it's very useful to to to keep this in mind um it can so it could like to boil my point down i guess the two sentences it can be productive doing this um it's just something that we really wanted to bring to the surface uh that we have to be aware of the distinction between the thing in the world and the mathematical formula that we're using to describe this thing in the awesome thanks for this summary and as people i think learn and read these papers they'll see that there are these expanding levels of analysis from like the one sentence don't confuse the map with the territory or what you just said there it can be productive you just have to raise this discussion to the surface and then there's more and more and more to uncover um kind of like chipping away at the base of the pyramid almost and finding out what are our assumptions and okay all these people criticizing elements of free energy principle or active inference what's their preference what do they want to see or are we all on the same aspirational team in that we all want to see a collaborative formal framework that also might help us with real world questions and also philosophical ones variously stephen then anyone else yeah i think this is this is a really good conversation and i think when i've heard other approaches say in the psychology fields which are kind of looking at how people do sense making they often outside of work with the free energy principle it tends to be energy gradients people you know what's route to least action or whether you notice the energy is lower so then you know how to act on the and of course the the ability to also bring in entropic kind of noisy statistical regularities which you wouldn't necessarily know why they're there um well you wouldn't know why they're there we no one knows they're there we can't even understand it there's something quite interesting and that seems to be um where to my understanding i don't know if there's any other models that are able to integrate the um the energy landscape with this kind of uh actual you know way to infer something plausibly from the entropic dynamics that are played out thanks steven blue then anyone else with a question so just to go back to what daniel was saying about incorporating the active inference framework do we want this really to be a framework that is concrete and definite and stable do we want to come to some kind of absolute end point here with with active inference or is this something that we want to be constantly evolving growing changing do we want to try try applying it like in several different seeing what works and what doesn't to then then move on so we have to think as we ask ourselves and as we you know discourage and encourage how the active inference framework is used what do we really want as scientists i think it's important to think about that great question blue really something we could all think a little bit about shannon anyone else sort of um following up but a little bit tangential what blue just mentioned um and maybe joe

can be here next week that would be great um but i'm wondering if we're having sort of an emerging folk active inference um so we have a like folk psychological intuitions about the world about how other people behave science uses slightly different maybe more complex intuitions to describe um how people perceive interact um to describe the psychology of individuals i'm wondering if this active inference concept is really great um how daniel talks about how we work in teams and how we're working together to resolve our uncertainty and we're coming together in this um this markov blanketed group of us compared to everyone who will be watching this live stream later and these are fruitful to making good team dynamics and to help learn and um some of us who aren't here right now are applying this to systems to business and organization and this is super useful it's not the goal of the scientists perhaps um but maybe we can bring this up again next week and just talk about how that's useful and how that's sort of emerging at the same time as these formalisms thanks shannon and i agree that um the team and the project and the mission sometimes sidestep or at least um ward off the philosophical questions it's like well let's finish this project with this data set we have get the publication or get this next checkpoint and then we'll see where we're at we'll have a new perspective on these eternal philosophical questions and so it won't be resolved in 2021 maybe not even in 2022 but we'll be working together and we'll be more productive and we'll have stabilized our informational niche our financial niche dave then anyone else dave dave he's still muted but okay okay anyone else want to uh add something uh i was just gonna say like something very briefly like um concerning what blue uh brought out which i think is um very fair i guess uh view of um a very fair perspective of you know the the current status of people that might choose to work on there is a good cluster i guess of people from different backgrounds that have some more or less shared goal that are ready to you know engage with each other and try i guess like to create something good with each other so if anything else that that's great i mean what else would you actually need to do science if not just a bunch of you know motivated people that get together and try to to work towards something altogether um something that you know might have um on the other hand push people away uh like at different stages in time was the the idea of promising something that was not really there yet in a way like the nobody would blame i guess anyone for for a project that is just in the workings for something that is evolving for something that is still you know that has still not reached its final form i guess the idea of reading you know a theory a unifying theory of this a unifying theory of that is what might have you know discouraged other people from from actually engaging with it so probably some sort of balance that i think is now in place is conducive to i

guess an even better research environment because i think more people can engage more people can see that there are you know constructive criticisms more people can see that you know it's not just uh you know a sort of environment completely detached from from reality there is there is a sense in which you know uh improvements happen when people engage and other people listen and you know propose new stuff which you know just a scientific method that's absolutely fine that's great i think we we had some very welcoming you know uh feedback from different like regardless of what we said which i thought you know is good for a healthy community nice i think yeah thank you yeah a mentor in writing told me that the three stages were the madman the architect and the interior designer so someone has to envision there could be a skyscraper here or there could be a stadium here and then it has to get planned out and interior design has to be kept in mind but it isn't actually applied and so not that those are the stages of a scientific theory but we're closer to the beginning and you're right that now there actually is enough to kind of take a bite out of and to build within and then for those who think that there's like little grounding or they want to make almost hyperbolic claims about how little is known um my questions would be first off relative to which body of literature that's ongoing and the second question would be have you read the 2007 spm textbook because that was 14 years ago and it demonstrates in a little bit like a textbook form which is more classical than the pre-print back and forth that's a lot of the basis and so there isn't any philosophy in that textbook but you'll find a lot of new approaches for thinking about cause and effect and dynamic systems modeling and what we can do with neuroimaging and behavioral data and heterogeneous data sets so awesome points um dave if you want to unmute and try again and then blue i'm i'm not sure why you're still muted there we go okay yep okay um one of the really crucial benefits of doing the deconflation that you are correctly insisting on is that the real essence of what's in scientific thinking or technical thinking or skill development is the analogies among different realms of thought realms of action if you conflate um entropy with the universal tendency you don't see you can't think about the actual differentiations of instruments that should that measure entropy cosmological ideas that discuss and structure our thinking about a different realm of activity and you certainly aren't going to see the analogy between those analogies but once you start pulling stuff apart then you can start doing mapping in detail and find what applications you can relate through the instrumental or realistic interpretation of any of these meta concepts you can start listing tables of correspondences you can make uh gadonkan experiment and you can make literally lab tests and draw out paradoxes where the prediction one wing of the metaphor looks like it

ought to align with another with the results in another wing and don't at that point now you can start thinking about what it is that we don't know yet so yeah yeah quick quick keep keep ripping the stuff apart when it's been conflated flogistan modeled a lot of real stuff and it was harmful the notion that things fall because they belong down that predicted a lot of things that modeled a lot of stuff and interfered with work that needed to be done thank you thanks steve nice points blue this is great because it just ties in everything that dave said but i really wanted to respond to manuel um so you know i don't necessarily have a perspective on whether we want active inference to be something solid and concrete like a law or something that's continually growing and evolving and so i i just pose it as like where do we really want it to to end um i think that um just like dave said you know there's always like we we had the laws of physics right like newtonian physics we had all these laws they were in place they worked for a long time and then quantum right so so i mean we don't know what is going to be underneath when we when we finally define an absolute there could be another layer that we just aren't seeing yet so i think that um you know just like newtonian physics and quantum physics we need to look at where the context is right and so there might be an active inference a a contextual barrier that we can say you know active inference is this version of active inference is applicable in this context and there might be some other deeper different version that applies in other contexts cool and i think that's partly how the the specific projects and deliverables and questions are what ground us like with dave's example about entropy yeah you can probably abstract entropy and start applying it to different domains and what would psychological entropy what would spiritual entropy be or you might be concerned with the car that you're building and you're like okay i have a preference for the more miles per gallon and less of this other emission and then entropy in an applied context is being pursued instrumentally with literal instruments and scientists are not just all-purpose thinkers when the scientist is on were instrumentalists and actually that makes me curious yella about one of your papers the anticipating brain is not a scientist so what was that about because the part in this paper about how the cognitive is in a unique position as the meta modeler reminded me a lot of this line of research so maybe could you summarize that or what was that paper about and how has that been developed and what you're exploring now oh great yeah um so uh yeah i think and can you hear me hey yeah i've got a comment um i only now realize that they used the scientist metaphor uh in both this paper and in in the in the old one uh and i think here it's very clear to distinguish between models and models of knowledge models and animals models and scientists models and they have different kind of properties in in the anticipating

brain paper we make the same distinction between the scientists model of the environment and an animal's model environment and there the distinction is more in terms of what are the the aims so to say of the model and where you might say well in the epistemic practice of science then the aim of the scientist model is to get an as accurate representation or inference or whatever from the outside world as possible so it is kind of a more epistemic process in which the ultimate aim is to to figure out something about the world and now transposing that question to the uh to the animal the problem is a different one it's not like the problem what's out there in the world is secondary to how can i make what's out there in a world compatible with the kind of existence that i am having or am and so there the epistemic story stands in the surface of something like flourishing or staying alive or homeostasis or so and there might be cases where actually and i think this is also related to the kind of the theory of action that's there like where uh inferring what's the exact state of the world is not actually the right thing to do but changing the world so as to make it conform to your expectations is the better way to do that so so that the kind of the epistemic aims in both settings the scientist setting and the animal in his environment setting are are what the the scientists in this everyday life setting are are slightly different and so that was also to counter the kind of very prevalent kind of helmholtzian models of perception that are there in the in the energy literature and and the processing literature where you say like well first and foremost a problem for an agent or any agent is to infer what's out there in the world and i think fep shows like no that that's that's a secondary problem the first problem is staying alive interesting and that's so funny because what do they say in legacy science publisher parish so it's also not thrive and communicate and mentor it's still phrased in a life or death way that's all this might be and then the eve wave understanding signs indeed that might be true yeah yes alliteration plus competition just irresistible any other thoughts otherwise there's a couple other points that would be awesome to kind of just in the last 20 minutes open up and then i'll re-upload this full video we can all think about some questions that we're having i guess one distinction that is clearly key from where you place it in the paper and the emphasis and everything is inference with versus inference within a model and this is gonna sound like a subtlety because it's two letters that you're appending to the word but they're two letters that make a big difference so what would be sort of the map territory distinction how do we think about inference with and within a model how do we speak clearly about different things that are happening who's going to take that one shall i okay okay good i think i i insisted on keeping in this distinction throughout all phases of the co-writing process so i guess i'm responsible for now also explaining it

coherently um i think the key distinction there is is to draw the distinction between a kind everyday wave understanding modeling of saying well a model can be used by something external to the model to model something that's even external to that right so i guess perhaps the notion of construal might actually also be something right there is there's there's there there's a scientist or there there's a practice or so that construes and relates a model uh uh to some target phenomenon so that you can can do a predictive analysis or or whatsoever uh and i think that's our everyday way of thinking about scientific um and then especially in the in the live as we know it papers at the first in 2014 uh paper something quite different is is happening right so so in most of the predicted processing literature this distinction that really made made that explicit but in 2013 that analysis is is not making clear how an agent or an animal can use a model to do something but shows makes a model and shows how somehow within that model a demarcation between agent and environment starts to emerge and because of the demarcation uh uh an inferential relationship so there's now there's not there's not a user of that model but somehow within a model their inferential relationship starts to starts to emerge and and so that's what we call the inference within a model the the emergence of the inferences within that model and the other one is something outside of the model is using the model to predict something or capture something that that's supposed to be this distinction maybe maybe we'll think about a a better a clearer semantic separation between those two concepts that should be could be a part cool in the four different kinds of i guess i want to say processes but the four different ways in which inference is sometimes deployed to be talking about generative processes and generative models you know the generative process is what actually the air moving through the house and the temperature as it's actually changing and then the generative model the predictive thermometer and there's other work that i know some of you have done about connecting systems that are probably less than lifelike and realizing that we can frame them in a cybernetic context using these um same constructs which is really an argument by demonstration almost about the philosophical points that you're trying to show it's like if somebody were making big claims about cybernetic systems and then someone said oh but it applies to a thermometer so either you need to pull back your claims a little bit or it's fine if you think thermometers have this attribute like regret that you're going to be studying but you don't get to say that the thermometer is of course insentient yet it's doing what you say is granting us sentience and these kinds of distinctions are not within active inference proper it's actually just about clarifying research directions and learning about how we work on these projects that go beyond the understanding of one person because there are a lot of technical

elements so kind of um remind us of that earlier thread which is how are we all going to learning about this and working within it and communicating within it when one could go deep into variational inference but you know i'm not going to be the one to find that there's some deep quirk or assumption and variational inference that invalidates some other technical piece of the puzzle so how do we almost kind of like the the doors in the titanic like separate different parts of our model so that we at least know if one of them is going to be unsupported at this moment or is it apt to even fail that the the there isn't a cascading failure for our actual direction but rather there's something that's modular that could be pulled out yeah i think there's lots of things that are lots of things about energy principle that are actually pretty trivial and some lots of things that are actually pretty uh speculative and high stakes and and indeed having some of the democrats that we make in our i guess allows you to separate some some empirical methods that are that are useful and and and kind of non non non-controversial uh separated indeed from from what we call boo uh the kind of blanket oriented ontology which is uh a big idea which we which we are kind of skeptical about uh at least i am i'm not sure i speak for all the uh uh uh co-authors but at least it has a different epistemic stages than the everyday bread and butter active inference that people are using and so separating those out a little bit uh using the different concepts and different assumptions that are made by each each approach uh probably also can give gives a better overview of the field and the things you need to buy into and don't necessarily need to buy into if you want to work with cool um dave then anyone else raising their hat yeah i'm going to ask a question not status or stature or number of papers what influence has the personality of carl fristen had on the development of these fields i want to go and break another lunch in in his favor um i haven't worked like directly with carl uh unless like on really small projects but he is just an extremely nice and kind person i think it is as simple as that like it is really hard not to to get along with them on some so i mean i want to be positive in a sense to say that it is really hard not to not to want to work with with such a nice person if we want to talk about that like trying like to stay away say from from his influence and his you know his status across fields it's just an incredibly nice person and even if sometimes you know there are some misunderstandings in communications there is i don't know uh there is a sense in which you feel like you want to work with that person i agree and also beyond niceness it is reflected through action and and mentorship and collaboration but also it's yes yellow and anyone else as well you know i can only uh uh second that i i spent some time in uh at ucl in in in this group i guess for for for outsiders and for brandings so you might think sometimes that that carl is able to speak a bit

uh too well right so listening to him if you don't really understand what he's saying it sounds it sounds enigmatic it sounds it sounds fascinating and it sounds it is but it also sounds incredibly clever right and so um uh i guess in terms of branding right what what what we see a lot now is that uh circle of people who really understand the stuff that's going on and and a group of people who are also put off by it a little bit because it it seems so intimidating from the outside and and uh i think that that's something that fpp as a whole kind of struggles with a little bit uh uh that it might seem a little bit esoteric if you're if you're not familiar uh uh uh with with with the keywords with haven't worked on that for for a while uh and i think that that's probably sparked but just these these kind of fantastic uh lectures that carl's able to get they're all over youtube and so and they're they're they're they're hard to get on the first go yeah and it's also notable that the 2013 single author paper life as we know it um it serves as a lightning rod for a lot of critique as does his 2010 paper uh unified brain theory and they're both i hesitate to call them old because they're less than you know a decade old in some cases but in terms of our current environment there's been a lot between then and now and also their single author and so let him speak and represent what he wants to share just like every author and collaboration and then there's a ton of nuance and a lot of development that's happened and i totally also resonate with the um criticism that many raise or at least the point that it is a rapidly developing field so it's like oh active inference can't deal with long time steps aha but then have you thought sophisticated active inference and so it does become a little bit of this sort of like criticism development criticism development which actually reminds me of daniel dennett's book darwin's dangerous idea and he says evolution by natural selection is the theory that's describing itself because it is improving itself and someone says well evolution doesn't include ecology or development or evolution okay well then we make ego evo and then we go from there and so there's sort of um a little bit of a blurriness between who's the detractor and who's actually out in front pulling the theory where it needs to be and that's also where cognitive diversity comes into play and just we're all single blue and then anyone else so just to kind of piggyback off of what you said i i always think um you know especially in our culture we have like this idea that there's a leader or a creator and in this situation it's carl fristin right so is there a leader or a creator of the field or has it become it's single-handedly invented a light bulb that's not how it happened but but we you know we kind of think this way and we've kind of placed first in in the leadership role here but but really i think that there's a lot going on underneath the curtains or underneath the blankets i agree and it's it's a meta problem with science communication like the 2018 wired article which the title be a

singular person and about something about their endocortex not this dedicated researcher has built a network and tools for 30 years that are inspiring the next generation researchers to make impact it doesn't fit quite as well in the headline and so it's like a communication and a meta communication question communicating about how these uh fields actually work and yeah there's going to be a lot of interesting times ahead on those frontiers i think any sort of closing thoughts or maybe we could go around and just give a last little thought or something that we're excited or curious about for next week's discussion when i'll fix it and we'll be able to include participants live also thanks manuel for for staying up late we'll have um martin beale on april 28th and that will be at um a better time i think it's two hours earlier than this one so stephen then anyone else yeah i'd just like to thank you for a really you know powerful presentation and answering a lot of difficult questions and also i suppose what i'll take away is this question about the blanket orientated ontology i think it's really good to have that